

Group 42 Progress Report: Category Detection for YouTube Videos

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1 Task Introduction

Our project builds a Natural Language Processing model that classifies the main category of a YouTube video. This task is significant because there are more content being uploaded on YouTube everyday, so classifying videos by category accurately can help filter content. The categories are the following:

- | | | |
|-------------------|--------------|----------------------|
| 1. Food/Cooking | 5. Education | 9. News |
| 2. Video Games | 6. Shopping | 10. Health & Fitness |
| 3. Sports & Games | 7. Vehicles | 11. Business |
| 4. Music | 8. Lifestyle | 12. Digital Media |

These categories are determined from YouTube’s category list (see here) and our team’s own generalization. Most importantly, it will be interesting to analyze the shared commonalities of video titles and descriptions across different categories. This task is challenging because most video titles do not explicitly state their category and may depend on the channel’s niche or current world events. To keep our labels consistent, we’ll write a short annotation guide with specific cues to look for, then have everyone in the team label the same starter set of headlines to line up our interpretations. Our goal is to reach agreement that’s clearly better than chance, while recognizing that some headlines will still be genuinely hard to classify.

2 Task Definition

Our dataset consists of YouTube video titles and descriptions, so each input example is a short piece of text. The task is a supervised multinomial classification problem: given a title and description, the model predicts the video’s primary category. We will use 11 classes, one for each of the categories mentioned above. We will stick with one label initially, as we are interested in seeing the primary category they are classified in.

3 Dataset

3.1 Plans for Collection

We will use an existing dataset *youtube-titles.txt* from <https://huggingface.co/datasets/AdamLucek/youtube-titles>. Each instance contains at minimum the video title (**video_title**) and a video description (**video_description**), and also metadata such as the `channel_name`, `channel_url`, `channel_id` and `video_id`.

We will use only publicly available data for this course project and follow the dataset's license/usage terms. We do not plan to perform additional web scraping. Prior to modeling, we will apply basic cleaning (removing emojis, links, promo codes and text normalization). We will also remove all columns other than the video title and description.

3.2 Expected Dataset Size

We expect to use $N = 1.5k$ examples. These are unlabeled, but we can use <https://tubepilot.ai/tools/youtube-category-checker/> to add labels for the remaining samples that our group didn't annotate by hand. We will create train/dev/test splits with an approximate 80/10/10 ratio (with the exact proportions determined by the outlet-based grouping constraint).

As required, we provide three example data points with labels assigned by our group:

1. **Title:** [Ludum Dare #43] The Resistance: Behind the Scenes
Description: "A game about batteries. Made in 48 hours for the Ludum Dare 43 game jam <https://ldjam.com/events/ludum-dare/43/the-resistance>. Support the creation of more tutorials and get early access to new videos: <https://www.patreon.com/SebastianLague>"
2. **Title:** Quadruple Stacked Ham Sandwich from Final Fantasy XV | Arcade with Alvin
Description: "This episode is sponsored by Squarespace. Head to <http://bit.ly/squarespacebabish> to save 10% off your first purchase of a website or domain using code BABISH. On this episode of Arcade with Alvin, Alvin makes the gravity-defying Stacked Ham Sandwich from Final Fantasy XV. Follow Alvin Zhou on Instagram: <https://www.instagram.com/alvin.zhou/> Pre-Order the new Basics with Babish Cookbook: <https://www.bingingwithbabish.com/cookbook> Babish Cookware on Amazon: <http://bit.ly/babishstore>"
3. **Title:** 7 tech leaders who ended up in prison
Description: "Yesterday, Sam Bankman-Fried was sentenced to 25 years in prison. Let's look at 7 other tech leaders and entrepreneurs who broke the law and went to prison. #tech #hacking #thecodereport Chat with Me on Discord <https://discord.gg/fireship> Resources Full Stripe Payments Course <https://fireship.io/courses/stripe-saas> Tech Trends 2024 <https://youtu.be/vyQv563Y-flk> Get More Content - Upgrade to PRO Upgrade at <https://fireship.io/pro> Use code YT25 for 25% off PRO access "

4 Team Contract

The team contract outlines the expectations of the team for the duration of the project. Submission of this project proposal implies that all members listed in the author section has read and agreed to the following contract.

4.1 Team Purpose

Our purpose is to apply the content and knowledge we learned during the SFWRENG 4NL3: Natural Language Processing class towards our project, classifying if a given news headline can indicate if the article is more right or left leaning.

4.2 Team Goals

- Complete all deliverables on time and to a high degree of quality.

- Maintain equal participation and contribution from all members
- Develop skills in collaboration, problem-solving, and accountability.
- Create an interesting project based on the course content we learn in class.

4.3 Communication Norms

Primary platform: Discord

Response expectation: Reply to discord messages within 24 hours.

Decision-Making: Strive for consensus when possible; else, majority vote will be used.

4.4 Ground Rules and Expectations

- Be respectful of all ideas and opinions.
- Come prepared for meetings.
- Complete/submit assigned work on time.
- It is up to the individual assigned to the task to inform the rest of the team if the task cannot be completed on time.
- All team members are expected to attend tutorial time [Wednesday, 4:30-5:20 PM], and to inform the team beforehand if they cannot make it.

4.5 Conflict Resolution

We recognize that conflicts may arise, particularly around workload distribution. To handle these situations:

1. Direct Conversation: The concerned member(s) will address the issue privately and respectfully with the person involved.
2. Group Discussion: If the problem continues, the whole team will discuss it openly and constructively, focusing on solutions (e.g., redistributing tasks, adjusting timelines).
3. Clear Expectations: If a team member repeatedly does not contribute, responsibilities will be formally reassigned, and the lack of contribution will be documented.
4. Escalation: If the issue remains unresolved, the team will notify the course instructor/TA with a written summary of the situation.

Signed by: Ahren Chen, Fei Xie, Grace Xiao